When Does Compressed KV Start Lying?

Token-Level Drift in KV Cache Compression

ExactKV Project
Technical Report (v3.0)

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Quantitative values from `reports/scale_7b/raw.json`, `reports/evidence_plus/raw.json`, `reports/external_panels/` *(including* `v30/`*), and* `reports/public_release/leaderboard_final.json`. *Claim boundaries:* `docs/CLAIM_BOUNDARIES.md`.

Abstract

Lossy KV-cache compression is widely deployed, but most evaluations report aggregate quality and stop there. ExactKV is a compressor-agnostic **crash-test framework** that measures first-divergence index, draft acceptance, and `exactkv_failure` under verifier-mediated semantics.

Across **8,132 completed GPU cells** (Llama-3.1-8B and Mistral-7B, five benchmark families, six built-in compressor classes[1]) all panels report `exactkv_failures=0`. Four main findings:

1. **Task type dominates.** `int4_sim`: 6% (MBPP code) → 11% (BFCL short) → 50% (BFCL long-gen) → **90%** (HF LongBench). H2O-style eviction at 75% kept: **100%** divergence on LongBench.
2. **Generation length scales within-task.** On BFCL, `int4_sim` rises from 9% (mnt=16) to 62% (mnt=256) — a 7× increase.
3. **Three distinct failure modes.** Top-k logit autopsy (1,103 divergent cells): near-tie noise (int8, rank 2.4, fdi=22); distribution shift (int4_sim, rank 3.5, fdi=8); attention destruction (H2O-style, rank 6.7, fdi=1). Three forensic case studies confirm each.
4. **100% downstream validity preserved.** Despite 50% token drift, ExactKV preserves all 106/106 full-KV valid BFCL tool calls (both models, all four task categories).

`int6_sim` (6-bit per-tensor) and `int4_per_vec_sim` (KIVI/KVQuant-style per-vector INT4, 2.06× int8 error) are GPU-validated non-catastrophic on both Mistral-7B and Llama-3.1-8B (v3.0, 1,568 cells total, `exactkv_failures=0`): both show 0% divergence on BFCL and MBPP; `int6_sim` reaches 37–47% and `int4_per_vec_sim` 56–57% on HF LongBench — per-vector granularity helps on structured tasks, while bit-width still matters at 8K LongBench context. Results are **not** benchmark scores, production claims, or a reproduction of VeriCache [8].

1 Introduction

Lossy KV-cache compression is widely deployed as a transparent optimization for LLM inference. Under greedy decoding, a single perturbed logit can flip the argmax — after which trajectories

[1] `noop`, `int8`, `int6_sim`, `int4_per_vec_sim`, `int4_sim`, and H2O-style eviction (`h2o_sim` family). `kivi_offline` is an external adapter diagnostic (Table 11), evaluated separately.

diverge. Aggregate metrics (perplexity, LongBench, task accuracy) average over this divergence and hide *where* it begins, *how severe* it is, and *which compressor design choices* drive it.

ExactKV reframes KV-cache evaluation around a single diagnostic question: **at what generated token does the compressed-cache path first stop matching full-KV greedy decoding?** We answer this with a compressor-agnostic crash-test framework that runs a draft/verify/commit loop and records first-divergence index, draft acceptance, verifier agreement, and exactness failures per cell.

Contributions:

1. A **compressor-agnostic crash-test framework** with leaderboard-style reporting for token-level drift, first-divergence index, acceptance rate, and exactness failures across compressors and models (§3–5).

2. **8,132 GPU cells** across Llama-3.1-8B and Mistral-7B, five benchmark families, six built-in compressor classes, with `exactkv_failures=0` throughout (§6).

3. Empirical evidence that **task type dominates drift** (6% code → 90% reading for `int4_sim`) and **generation length scales it** (9% → 62% on BFCL, 7×) (§6.4–6.12).

4. A **logit autopsy** over 1,103 divergent cells identifying three mechanistically distinct failure modes: near-tie noise (int8), distribution shift (int4_sim), and attention destruction (H2O eviction) (§10, §15).

5. **Downstream validity measurement**: despite 50% token drift, all 106/106 full-KV valid BFCL tool calls are preserved (§6.11).

6. **GPU validation** of `int6_sim` and `int4_per_vec_sim` confirming non-catastrophic drift on structured tasks; per-vector granularity helps on BFCL/MBPP while bit-width still matters at 8K LongBench context (Sections 13–14, Table 25).

1.1 Shared problem framing with VeriCache

VeriCache [8] and ExactKV share the observation that lossy KV may look acceptable on short or aggregate metrics but **diverges more as decoding continues**, and catastrophic for code and tool-calling. VeriCache uses compressed KV to **draft** and full KV to **verify/correct**, guaranteeing **identical greedy output** to full-KV inference.

1.2 Where ExactKV diverges from VeriCache

VeriCache is a **serving/system** paper. Its contribution is not merely "draft then verify." It makes that loop practical by: (1) keeping compressed KV on GPU, (2) keeping full KV in CPU/storage, loading only for verification, (3) **cross-resource staggering** (HBM-bound draft vs interconnect-bound verify), (4) reporting serving throughput (up to ∼4× vs full-KV in their evaluation) with identical outputs.

ExactKV is an **evaluation crash-test**. It measures first divergence, acceptance, and exactness failures, *not* deployment throughput. It does *not* reproduce VeriCache scheduling or memory tiering.

#	Finding	Key number
1	Task type dominates drift	int4_sim: MBPP 6% → BFCL 11% → BFCL long 50% → LongBench 90%
2	Generation length scales drift	int4_sim mnt=16: 9% → mnt=256: 62% (7×)
3	Eviction > quantization	H2O-style 75% kept → 100% LongBench divergence
4	Three distinct failure modes	int8: near-tie (rank 2.4, fdi=22); int4_sim: shift (rank 3.5, fdi=8); H2O: destruction (rank 6.7, fdi=1)
5	100% downstream validity	ExactKV preserves 106/106 valid BFCL tool calls despite 50% drift
6	Zero correctness failures	`exactkv_failures=0` across all 8,132 GPU cells

Table 1: Main findings summary.

1.3 Main Findings at a Glance

2 Definitions

2.1 Reference paths

- **Full-KV reference:** greedy generation with uncompressed KV.
- **Lossy path:** compressed-KV greedy without verifier.
- **ExactKV path:** verifier-mediated draft/verify/commit (`ExactKVGenerator`).

2.2 Metrics (formal)

Table 2: Core metrics (formal definitions).

Metric	Definition
`first_divergence_index`	First position where the *lossy* token ≠ full-KV greedy (before correction).
`acceptance_rate`	`total_accepted / total_drafted` across verification rounds.
`exactkv_failure`	Final ExactKV output ≠ full-KV greedy (verifier/commit failure).
`divergence_rate`	Fraction of cells with lossy-path divergence.
`divergence_stability`	1 − `divergence_rate` (complement).
`compression_ratio`	Stored tensor byte ratio only (not active GPU memory).

Corrections vs rejections: `total_rejected` counts rejected *draft tokens*, `total_corrections` counts verification *rounds* with a correction. For `int4_sim`: 1284/1502 accepted, 218 rejected tokens, **116 correction rounds**.

2.3 Algorithm

```
Prefill -> FullKVState + CompressedKVState
while generated < max_new_tokens:
  DRAFT from compressed KV (up to draft_len tokens)
  VERIFY each token vs full-KV greedy
  COMMIT accepted prefix, correct on mismatch
```

```
  ALIGN compressed state from authoritative full KV
return ExactKV output_ids
```

Greedy decoding only in this release.

3 Method

Each **cell** is (model, compressor, prompt, `max_new_tokens`) with `draft_len=4`, `seed=0`. Three modes per cell: **full**, **lossy**, **exactkv**. Leaderboard score: $0.35\,a + 0.25\,v + 0.20(1 - \hat{d}) + 0.10(1 - f) + 0.10\,s$.

4 Experimental Setup

Table 3: Headline release panel (`reports/scale_7b/raw.json`).

Parameter	Value
Panel ID	`phase_a_unified_panel`
Total cells	**1500** $= 2 \times 5 \times 50 \times 3$
Models	Llama-3.1-8B (750), Mistral-7B (750)
Stack	torch 2.8.0+cu128, transformers 5.12.1
`draft_len`	4 `max_new_tokens` $\in \{4, 8, 16\}$
Decoding	Greedy only
Prompts	50 variants from 4 stress templates (capital, math, JSON, code)
Categories	capital_france 390, simple_math 390, json_tool 360, code_fn 360
Execution	Sequential per model, `deterministic_mode=false`

Validated compressors: `noop`, `int8`, `int4_sim`. Real KIVI/KVQuant/SnapKV/TurboQuant are not in the headline panel. The raw JSON also contains proxy/probe slots (`spectralquant`, `shard`) excluded from all analysis; see Limitations (§22).

1,560 GPU cells total: 216 Llama-3.1-8B cells across bundled LongBench/RULER/BFCL/HumanEval pilots; 144 MBPP cells across both models (`mbpp_gpu_raw.json`); 1,200 BFCL export-50 tool-call drift cells across both models — `exactkv_failures=0` throughout. Not official benchmark scores.

5 Results: Validated Compressors

Table 4: built-in compressors with direct ExactKV integration (`noop`, `int8`, `int4_sim`). Aggregates from `compressor_summary` in `reports/scale_7b/raw.json` (300 cells each: 150 per model).

`int8`/`noop`: no lossy divergence. `int4_sim`: drifts in 52% of cells yet `exactkv_failure=0` because the verifier corrects.

Leaderboard-style scores (formula: $0.35a + 0.25v + 0.20(1 - \hat{d}) + 0.10(1 - f) + 0.10s$, source: `reports/public_release/leaderboard_final.json`):

5.1 Metric interpretation

Divergence rate and acceptance rate measure *different* failure modes. Low lossy-path divergence does not imply high verifier acceptance (and vice versa). Example: `int4_sim` at

Compressor	Cells	Accept.	Div.†	Stab.	1st div.§	Fail.
noop	300	1.000	0.00	1.00	n/a	0.00
int8	300	1.000	0.00	1.00	n/a	0.00
int4_sim	300	0.851	0.52	0.48	2.0	0.00

Table 4: Built-in compressors from `compressor_summary`. †Lossy-path divergence fraction. §Mean over divergent cells only. Histogram for `int4_sim`: index 1→78, 3→39, 5→24, 8→13 cells.

Compressor	Model	Score	Accept.	Div. rate†	Stab.	Cells
noop	Llama-3.1-8B	1.000	1.000	0.000	1.000	150
int8	Llama-3.1-8B	1.000	1.000	0.000	1.000	150
noop	Mistral-7B	1.000	1.000	0.000	1.000	150
int8	Mistral-7B	0.983	1.000	0.000	0.827	150
int4_sim	Llama-3.1-8B	0.859	0.852	0.520	0.480	150
int4_sim	Mistral-7B	0.851	0.837	0.507	0.493	150

Table 5: Leaderboard-style scores for validated built-in compressors. †Div. rate = raw lossy-path token divergence fraction (same metric as Table 4). `int8` Mistral: divergence_rate=0.000 (zero cells diverge); `Stab.`=0.827 is the **leaderboard stability subscore** from `leaderboard_final.json`, which is distinct from the formal metric divergence_stability = 1−divergence_rate (Table 2); under the formal metric, `int8` Mistral divergence_stability=1.000. `spectralquant` (MOCK→`int4_sim`) and `shard` (PROBE_ONLY) excluded from ranked table.

`max_new_tokens=16` can show first divergence at index 8 with acceptance 0.94 — how late drift occurs matters.

5.2 Evidence-plus long-context panel

Source: `reports/evidence_plus/raw.json` (RunPod A5000, June 2026). 144 cells = $2 \times 6 \times 2 \times 3 \times 2$, prefill buckets 512/1024, `max_new_tokens` $\in \{16, 32\}$, built-in compressors only. By context

Compressor	Cells	Accept.	Div. rate	Fail.
noop	48	1.000	0.00	0.00
int8	48	1.000	0.00	0.00
int4_sim	48	0.985	0.31	0.00

Table 6: Evidence-plus aggregates. `int4_sim`: Mistral 41.7%, Llama 20.8% divergence (24 cells each).

bucket (all compressors): 512→11.1% divergence, 1024→9.7%. Diagnostic timing: mean 3854 ms, p90 5178 ms per cell (not serving throughput).

5.3 External benchmark smoke panels

Source: `analysis_pack.json` (216 Llama-only), `mbpp_gpu_raw.json` (144 cells). `exactkv_failures=0` on all panels. 216-cell subset: `noop`/`int8`: 0%; 15 divergent cells (`int4_sim` only).

Panel	Source	Model	Ctx (K)	MNT	Cells	Div.	Acc.	Fail.	P90 ms
LongBench	pilot	Llama-3.1-8B	2,4	16,32	72	0.083	0.994	0	8781
RULER 2K–4K	pilot	Llama-3.1-8B	2,4	16,32	48	0.083	0.993	0	8793
RULER 8K	pilot	Llama-3.1-8B	8	16,32	24	0.083	0.993	0	13640
BFCL	pilot (4)	Llama-3.1-8B	1,2	16,32	48	0.063	0.999	0	6342
HumanEval	pilot (4)	Llama-3.1-8B	1,2	32	24	0.000	1.000	0	6392

Table 7: Llama-3.1-8B external smoke panels (216 cells, `noop`/`int8`/`int4_sim`). Drift metrics; not official scores.

Panel	Source	Models	Ctx (K)	MNT	Cells	Div.	Acc.	Fail.	P90 ms
MBPP	pilot (6)	Llama+Mistral	0.5,1	16,32	144	0.021	0.999	0	5121

Table 8: MBPP code-drift smoke (144 cells, `noop`/`int8`/`int4_sim`). Not pass@1; no test execution.

Compressor	Model	Cells	Div. rate	CI_{95} low	CI_{95} high
`noop`	Llama-3.1-8B	200	0.000	0.000	0.019
`noop`	Mistral-7B	200	0.000	0.000	0.019
`int8`	Llama-3.1-8B	200	0.000	0.000	0.019
`int8`	Mistral-7B	200	0.000	0.000	0.019
`int4_sim`	Llama-3.1-8B	200	0.055	0.031	0.096
`int4_sim`	Mistral-7B	200	0.170	0.124	0.228

Table 9: BFCL export-50 tool-call drift panel (1,200 cells, both models, 50 prompts). `exactkv_failures=0`. Source: `bfcl_export_50_raw.json`. Drift measurement only (not JSON-completeness); `max_new_tokens`$\in \{16, 32\}$. Mistral-7B `int4_sim` divergence (17.0%) is 3× Llama-3.1-8B (5.5%).

Notable: LongBench `int4_sim` 25%; RULER 8192 `niah_single` 33%; BFCL pilot `int4_sim` 18.75%; BFCL export-50 `int4_sim` 11.3% (Llama 5.5%, Mistral 17.0%); HumanEval zero drift; MBPP 3/144 divergent. `int8`/`noop`: 0% divergence on all panels.

Panel	n	Div.	Rate	CI_{95} low	CI_{95} high
LongBench pilot	72	6	8.3%	3.9%	17.0%
RULER 2K–4K	48	4	8.3%	3.3%	19.7%
RULER 8K	24	2	8.3%	2.3%	25.8%
BFCL pilot	48	3	6.3%	2.2%	16.8%
HumanEval pilot	24	0	0.0%	0.0%	14.2%
MBPP pilot	144	3	2.1%	0.7%	6.0%
BFCL exp-50 `int4_sim` Llama	200	11	5.5%	3.1%	9.6%
BFCL exp-50 `int4_sim` Mistral	200	34	17.0%	12.4%	22.8%
BFCL exp-50 `int8`/`noop`	800	0	0.0%	0.0%	0.5%
All 216 ext. cells	216	15	6.9%	4.3%	11.1%

Table 10: Wilson 95% CIs on divergence rate (panel-level). Source: `scripts/build_external_analysis_pack.py`. "All 216 ext. cells" refers to the **initial** LongBench/RULER/BFCL/HumanEval pilot panel (first workflow, Llama-only); the full 1,560-cell external smoke total includes MBPP and BFCL export-50.

Overall acceptance (initial 216 ext. pilot cells): Full-acceptance rate 93.7%, Wilson 95% CI [90.5%, 96.8%]. Source: `analysis_pack.json` → `totals.acceptance_full_rate_ci95`.

5.4 KIVI offline compressor panel — real HF results (640 cells)

Source: `kivi_longbench_hf_raw.json` (320 cells) + `kivi_mbpp_hf_raw.json` (320 cells). RunPod RTX A5000, June 2026. `exactkv_failures=0` on all 640 cells.

Compressor	LB div.	LB acc.	MBPP div.	MBPP acc.	Fail.
`noop`	0.0%	1.000	0.0%	1.000	0
`int8`	13.8%	0.994	0.0%	1.000	0
`int4_sim`	91.2%	0.818	5.0%	0.995	0
`kivi_offline`	**100.0%**	**0.004**	**100.0%**	**0.001**	**0**

Table 11: KIVI offline compressor panel (80 cells each × 4 compressors × 2 datasets). `kivi_offline` produced catastrophic **adapter-level** drift in this ExactKV integration (near-zero acceptance). Because this is an offline simulate-path adapter rather than KIVI production CUDA/Triton serving, these results diagnose the adapter/integration path, *not* the KIVI algorithm as deployed. ExactKV detected and corrected all cells (`exactkv_failures=0`). Real HF `int4_sim` drift (LB: 91.2%) greatly exceeds bundled pilot (20.8%).

6 ExactKV-HF-LongBench Drift Panel (v2.6 — Complete)

Status: Complete. 720 cells, both models, `exactkv_failures=0`.

Key finding (task-type sensitivity): Real HF LongBench open-text reading/summarization at 2K–8K prefill produces **very high int4_sim divergence** (90.4%), the highest seen across all

Model	Compressor	n	Div.	Rate	CI_{95}	Mean acc.	EKV fail
Llama-3.1-8B	noop	120	0	0.0%	[0.0%, 3.1%]	1.000	0
Llama-3.1-8B	int8	120	33	27.5%	[20.3%, 36.1%]	0.988	0
Llama-3.1-8B	int4_sim	120	110	**91.7%**	[85.3%, 95.4%]	0.825	0
Mistral-7B	noop	120	0	0.0%	[0.0%, 3.1%]	1.000	0
Mistral-7B	int8	120	26	21.7%	[15.2%, 29.9%]	0.988	0
Mistral-7B	int4_sim	120	107	**89.2%**	[82.3%, 93.6%]	0.861	0
All (int4_sim)	all	**240**	**217**	**90.4%**	**[86.1%, 93.5%]**	0.843	**0**

Table 12: ExactKV-HF-LongBench v2.6 (720 cells, 10 real HF subsets, contexts 2K/4K/8K, `mnt`=32/64, both models). int4_sim: **90.4%** divergence on long-context open-text tasks — highest across all ExactKV panels. int8 shows 24.6%, unlike its 0% on BFCL/MBPP/RULER. `exactkv_failures=0`. Median first-div index: 4 (Llama) / 6 (Mistral).

ExactKV panels. Even int8 shows 24.6% divergence on LongBench tasks, unlike its 0% on BFCL, MBPP, and RULER. Divergence occurs very early (median token 4–6 out of 32–64 generated). `exactkv_failures=0` across all 720 cells.

Table 13 shows per-subset int4_sim divergence. `lcc` (code completion) is notably lower for Mistral (42%), consistent with code tasks being more stable under int4_sim across panels.

Subset	Type	Llama int4_sim	Mistral int4_sim
2wikimqa	multi-hop QA	12/12 (100%)	12/12 (100%)
gov_report	summarization	11/12 (92%)	12/12 (100%)
hotpotqa	multi-hop QA	12/12 (100%)	12/12 (100%)
lcc	**code completion**	**11/12 (92%)**	**5/12 (42%)**
multifieldqa_en	multi-field QA	6/12 (50%)	10/12 (83%)
narrativeqa	reading comp.	12/12 (100%)	12/12 (100%)
passage_retr.	doc. retrieval	12/12 (100%)	12/12 (100%)
qasper	academic QA	12/12 (100%)	12/12 (100%)
samsum	dialogue summ.	10/12 (83%)	12/12 (100%)
trec	classification	12/12 (100%)	8/12 (67%)

Table 13: Per-subset int4_sim divergence in HF LongBench v2.6 (2K/4K/8K context, `mnt`=32/64). `lcc` (code completion) shows lower divergence for Mistral (42%) vs. reading-comprehension tasks (100%), consistent with code-task stability across all ExactKV panels. `exactkv_failures=0` for all 720 cells.

7 BFCL Tool-Call Validity Panel (v2.7 — Complete, 1,200 cells)

The 48-cell BFCL pilot used `max_new_tokens` $\in$ `{16, 32}`, too short for complete JSON tool calls. A dedicated 1,200-cell validity panel (`max_new_tokens=128/256`) was completed with both models.

Compressor	Model	n	Div.	Rate	EKV fail
noop	Llama	200	0	0.0%	0
noop	Mistral	200	0	0.0%	0
int8	Llama	200	0	0.0%	0
int8	Mistral	200	3	1.5%	0
int4_sim	Llama	200	90	**45.0%**	0
int4_sim	Mistral	200	111	**55.5%**	0
All (1,200)	both	1,200	204	**17.0%**	**0**

Table 14: BFCL tool-call validity panel v2.7 (1,200 cells, mnt=128/256, both models). `int4_sim` diverges 45.0% on Llama and 55.5% on Mistral with long generation, vs 11% at mnt=16/32 — confirming generation length as a primary driver. `int8` is near-zero. `exactkv_failures=0` throughout. Artifact: `bfcl_validity_v27_merged_raw.json`.

8 H2O-Style Token-Eviction Panel (v2.8 — Complete, 800 cells)

Status: Complete. 800 cells, 5 eviction variants, both models, `exactkv_failures=0`. Artifact: `h2o_v28_merged_raw.json`.

Eviction mechanism: `h2o_sim_75`, `h2o_sim`, and `h2o_sim_25` keep approximately 75%, 50%, and 25% of prefill tokens using a simplified heavy-hitter-style eviction heuristic (cumulative attention score across heads). Lowest-scoring tokens are evicted after the full prefill; retained tokens include attention sinks (positions 0–3) plus the highest-attention tokens by recency/score. This approximates the H2O [9] eviction policy but omits the original online streaming update and GPU-efficient sparse-attention kernel. Results characterize the eviction compressor *class*, not the published H2O system.

Key result: Even mild eviction (75% of tokens kept) causes **100% divergence** on LongBench reading tasks for both models. The H2O-style simulation reaches mean lossy rank 6.5–6.7 at divergence (vs. rank 3.5 for `int4_sim`), confirming attention destruction as a qualitatively distinct failure mode. `exactkv_failures=0` across all 800 cells.

9 Cross-Panel Divergence Analysis: Task-Type Sensitivity

Table 15 summarises int4_sim, int8, and noop divergence rates across all completed ExactKV panels, revealing task type as the dominant factor.

Panel	Task	Ctx	mnt	Models	noop	int8	int4_sim
Headline (v1)	mixed	∼500	16/32	L+M	0.0%	0.0%	**51.3%**
Evidence-plus (v2)	mixed	512/1024	16	L+M	0.0%	0.0%	**31.2%**
MBPP code-drift	code	512/1024	16/32	L+M	0.0%	0.0%	**6.2%**
BFCL export-50	tool-call	1K–2K	16/32	L+M	0.0%	0.0%	**11.2%**
BFCL validity v2.7	tool-call	1K–2K	128/256	L+M	0.0%	0.8%	**50.3%**
HF LongBench v2.6	reading/summ.	2K–8K	32/64	L+M	0.0%	24.6%	**90.4%**

Table 15: Cross-panel int4_sim/int8/noop divergence rates. noop=0% throughout confirms correct baseline. Task type dominates: code (6%) < tool-call short-gen (11%) < tool-call long-gen (50.3%) < reading (90%). int8 is zero on BFCL export-50/MBPP/RULER, near-zero on BFCL validity (0.8%), and substantial only on LongBench (24.6%). Generation length also matters (BFCL 11%→50% as mnt 16→256). `exactkv_failures=0` across all panels. L=Llama-3.1-8B, M=Mistral-7B.

10 Mechanistic Analysis: Divergence Autopsy

The cross-panel task-type hierarchy is supported by a token-level mechanistic analysis. ExactKV stored top-5 full-KV and lossy logit distributions at each divergence point (`--store-top-k-logits`), enabling a logit autopsy across 1,103 divergent cells from v2.6, v2.7, and v2.8 panels.

Compressor	n div	top-1 flip	near-tie	mean margin	mean rank	med. fdi
int8	62	69%	**66%**	0.116	2.4	22
int4_sim	563	83%	26%	0.305	3.5	8
h2o_sim_75	160	**100%**	0%	0.545	6.5	1
h2o_sim	160	**100%**	0%	0.536	6.7	1
h2o_sim_25	158	**100%**	1%	0.551	6.5	1

Table 16: Logit autopsy across 1,103 divergent cells. near-tie: full-KV margin < 0.05. fdi = first-divergence index. Three distinct failure modes: (1) *near-tie noise* (int8), (2) *distribution shift* (int4_sim), (3) *attention destruction* (H2O-style). `exactkv_failures=0` across all 1,103 cells.

Three qualitatively distinct failure modes emerge:

(1) Near-tie noise (int8): 66% of divergences are near-ties (margin < 0.05). Lossy mean rank 2.4; late divergence (fdi=22). Small 8-bit noise flips only closely-contested decisions — explaining why int8 diverges on LongBench (uncertain outputs) but not structured tasks.

(2) Distribution shift (int4_sim): 83% top-1 flip, 26% near-tie, mean margin 0.305. The full-KV model is moderately confident, yet int4_sim selects ∼3rd–4th ranked token (mean rank 3.5), diverging at median token 8. Compression biases the activation toward plausible but incorrect alternatives.

(3) Attention destruction (H2O-style): 100% top-1 flip, 0% near-tie, mean margin 0.54, median fdi=1. Eviction eliminates context anchors from the very first generated token; the lossy model selects tokens ranked 6th–7th in the full-KV distribution.

These three modes directly explain the task-type hierarchy: near-tie noise only surfaces on long-context uncertain tasks (LongBench); distribution shift affects all tasks but worsens with longer generation; attention destruction is immediate regardless of task. `exactkv_failures=0` across all 1,103 divergent cells.

11 Downstream Task-Impact Metrics

Token-level drift is ExactKV's primary metric, but downstream task output validity is the ultimate concern. We report tool-call validity preservation from BFCL v2.7.

Compressor	Model	n	Drift	Full-KV valid	ExactKV valid	Preserved
noop	Llama	200	0.0%	63/200 (31.5%)	63/200 (31.5%)	**100%**
noop	Mistral	200	0.0%	43/200 (21.5%)	43/200 (21.5%)	**100%**
int8	Llama	200	0.0%	63/200 (31.5%)	63/200 (31.5%)	**100%**
int8	Mistral	200	1.5%	43/200 (21.5%)	43/200 (21.5%)	**100%**
int4_sim	Llama	200	45.0%	63/200 (31.5%)	63/200 (31.5%)	**100%**
int4_sim	Mistral	200	55.5%	43/200 (21.5%)	43/200 (21.5%)	**100%**

Table 17: BFCL tool-call validity preservation (v2.7, 1,200 cells). Despite 45–55% token-level drift for `int4_sim`, ExactKV preserves **100%** of full-KV valid tool calls for both models. *Note: "preservation" means ExactKV matches full-KV output, not that full-KV itself always produces valid calls. The full-KV baseline achieves only 26.5% (106/400) valid tool calls on this panel.*

Category	n	Drift	Full-KV valid	ExactKV valid	Preserved
simple	104	49.0%	38/104 (37%)	38/104 (37%)	**100%**
parallel	104	54.8%	31/104 (30%)	31/104 (30%)	**100%**
multi_turn	104	59.6%	24/104 (23%)	24/104 (23%)	**100%**
ast_eval	88	35.2%	13/88 (15%)	13/88 (15%)	**100%**

Table 18: BFCL downstream validity by task category (`int4_sim`, 400 cells). ExactKV achieves 100% preservation across all four categories despite 35–60% drift.

Task	int4_sim div	int4_sim acc	int8 div	int8 acc
trec	83.3%	0.904	0.0%	1.000
multifieldqa_en	66.7%	0.955	16.7%	0.997
lcc (code)	66.7%	0.908	4.2%	0.994
qasper	100.0%	0.899	16.7%	0.994
samsum	91.7%	0.854	16.7%	0.993
gov_report	95.8%	0.781	16.7%	0.997
hotpotqa	100.0%	0.787	37.5%	0.972
2wikimqa	100.0%	0.880	29.2%	0.987
narrativeqa	100.0%	0.716	66.7%	0.973
passage_retrieval	100.0%	0.746	41.7%	0.974

Table 19: LongBench per-token acceptance rate as draft-utility proxy. Even when all cells diverge (100%), `int4_sim` achieves 72–95% per-token acceptance: ExactKV speculatively uses the lossy draft for most generation before correcting the divergent suffix. `int8` reaches 97–100% on the same tasks.

12 Scaling Analysis: Context-Length and Generation-Length

Compressor	2K context	4K context	8K context
noop	0.0%	0.0%	0.0%
int8	21.2%	27.5%	25.0%
int4_sim	**95.0%**	**86.2%**	**90.0%**

Table 20: Context-length scaling on HF LongBench v2.6 (80 cells each). `int4_sim` is near-ceiling at all lengths — task type dominates, not context length. `int8` is moderate (21–28%) and roughly flat.

Compressor	mnt=16	mnt=32	mnt=128	mnt=256
noop	0.0%	0.0%	0.0%	0.0%
int8	0.0%	0.0%	0.5%	1.0%
int4_sim	9.0%	13.5%	**38.5%**	**62.0%**

Table 21: Generation-length scaling on BFCL tool-calling (both models combined). `int4_sim` divergence scales 7× from mnt=16 to mnt=256. `int8` remains near-zero throughout. Generation length is the dominant driver of BFCL divergence.

13 int6_sim: Non-Catastrophic Intermediate Compressor

`int6_sim` implements 6-bit symmetric quantisation simulation (64 discrete levels, scale = $\max(|x|)/31$). It fills the quantisation-error gap between `int8` and `int4_sim`. GPU-validated (v3.0, both models): 0% BFCL/MBPP; 37.5%/47.2% LongBench (Mistral/Llama). Mean absolute error: 4.15× `int8` and 0.23× `int4_sim`.

Compressor	bits	levels	ratio vs fp16	quant. error	type
int8	8	256	0.50×	1.0× (baseline)	real
int6_sim	**6**	**64**	**0.375**×	**4.15**× int8	sim
int4_sim	4	16	0.25×	18.3× int8	sim

Table 22: `int6_sim` compression curve. GPU-validated (v3.0, both models): 0% BFCL/MBPP; 37.5% (Mistral) / 47.2% (Llama) LongBench. `exactkv_failures=0`.

14 int4_per_vec_sim: Per-Vector INT4 Quantisation

`int4_per_vec_sim` implements per-vector symmetric INT4 quantisation, inspired by the per-vector/per-channel granularity used in KVQuant/KIVI-style KV quantisation [2, 6]. Instead of a single scale over the entire KV tensor, each token-position vector of size `head_dim` receives its own scale: scale $= \max(|x|)/7$ computed independently for each `[batch, head, seq_pos]` vector. This eliminates cross-head quantisation contamination from outlier attention heads.

Quantisation error comparison on a realistic KV cache shape [1, 32, 512, 64] with simulated outlier heads (first 4 heads 5× larger activations):

Compressor	Granularity	MAE	×int8	Notes
int8	per-tensor	0.067	1.00×	Real compressor
int4_per_vec_sim	**per-vector**	**0.138**	**2.06**×	**KVQuant/KIVI-style**
int6_sim	per-tensor	0.275	4.10×	Intermediate
int4_sim	per-tensor	0.844	12.59×	Per-tensor catastrophic

Table 23: Per-vector INT4 achieves 6× lower error than per-tensor INT4 on realistic KV shapes with outlier heads (2.06× int8 vs 12.59× for per-tensor). Granularity dominates bit-width: `int4_per_vec_sim` (4-bit) beats `int6_sim` (6-bit, per-tensor).

Predicted vs observed v3.0 divergence (both models, task-family means):

Task family	int8	int4pv pred.	int4pv obs.	int6 obs.	int4 obs.
MBPP code	0%	∼0–2%	**0%**	0%	6.3%
BFCL tool-call	0%	∼1–4%	**0%**	0%	52.5%
HF LongBench	18.1%	∼30–50%	**56.3%**	42.4%	85.4%

Table 24: Analytical prediction vs GPU-observed divergence for `int4_per_vec_sim`. v3.0 panel (both models, `exactkv_failures=0`). LongBench observed value is slightly above the predicted range but remains between `int8` and per-tensor `int4_sim`.

Key insight: The per-vector vs per-tensor distinction reveals that quantisation *granularity* is at least as important as bit-width for KV cache compression on structured tasks; at 8K context, bit-width still contributes meaningfully.

Compressor	MBPP	BFCL	LongBench	Class
int8	0%	0%	18.1%	8-bit quant
int6_sim	0%	0%	42.4%	6-bit quant
int4_per_vec	0%	0%	56.3%	4-bit per-vector
int4_sim	6.3%	52.5%	85.4%	4-bit per-tensor
h2o_sim_75	—	—	100%	Eviction (75% kept)

Table 25: Core benchmark curve (both-model means; v3.0 quant compressors; H2O from v2.8). `exactkv_failures=0` throughout.

15 Forensic Divergence Case Studies

Three representative cells from the logit autopsy (Section 10) illustrate the mechanistic failure modes in detail.

15.1 Forensic Case A — int8: Near-tie noise

Rank	Token	Full-KV prob	Lossy prob	Note
1	`' very'`	0.129	0.113	full-KV argmax
2	`' fine'`	0.113	0.129	**int8 chose this**
3	`' pretty'`	0.083	0.082	
4	`' great'`	0.075	0.073	
5	`' good'`	0.065	0.064	

Table 26: int8, NarrativeQA, 4K context, token 53. Margin 0.016 (near-tie <0.05). INT8 noise flips `' very'` → `' fine'` — a near-synonym. `exactkv_failure=0`.

15.2 Forensic Case B — int4_sim: Distribution shift

Setting: NarrativeQA reading comprehension, 2K context, Llama-3.1-8B. The full-KV model is moderately confident (margin 0.158), yet `int4_sim` selects a rank-4 token. This is the distribution-shift signature: 4-bit quantisation biases attention activations so that activation energy accumulates on a plausible but incorrect continuation, even when the full-KV model's preference is clear. Divergence occurs at token 1 (the very first generated token), meaning the entire generation trajectory is corrupted from the start.

Rank	Token	Full-KV prob	Note
1	`','`	0.271	full-KV argmax
2	`' they'`	0.232	
3	`' and'`	0.202	
4	`' I'`	0.113	**int4_sim chose this (rank 4)**
5	`' it'`	0.039	

Table 27: int4_sim, NarrativeQA, 2K context, token 1. Margin 0.158 (full-KV confident). 4-bit quantization shifts attention activation toward rank-4 token. `exactkv_failure=0`.

15.3 Forensic Case C — H2O-style: Attention destruction

Setting: HotpotQA multi-hop reasoning, 2K context, Llama-3.1-8B, h2o_sim (50% tokens kept). This is qualitatively different from both Cases A and B: the H2O-style eviction discards the factual anchor tokens needed for multi-hop reasoning, leaving only attention-sink tokens and the recency window. The result is not a near-tie or a biased distribution — it is a *collapsed representation* where the evicted model selects a token that does not even appear in the full-KV top-5. The fact that the full-KV model distributes probability across numeric tokens (suggesting factual recall) while H2O selects a question mark (suggesting confusion) confirms total context destruction.

Rank	Token	Full-KV prob	Note
1	`'1'`	0.056	full-KV argmax
2	`'15'`	0.039	
3	`'20'`	0.035	
4	`'10'`	0.034	
5	`'5'`	0.034	
>5	`'?'`	<0.034	**H2O-style chose this (rank >5!)**

Table 28: h2o_sim (50% kept), HotpotQA, 2K context, token 1. H2O-style chose outside top-5 entirely — the evicted tokens included factual anchors. Collapsed representation; not a near-tie or distribution shift. `exactkv_failure=0`.

16 Divergence Case Studies (Release Panel)

Case	Compressor	1st	Acc.	Corr.	Lossy vs full snippet
A	`int4_sim`	3	0.75	1	`art` vs `many`
B	`int4_sim`	1	0.50	1	math drift at token 1
C	`int4_sim`	5	0.70	1	JSON: `temperature` vs `country`
D	`int4_sim`	8	0.94	1	late divergence at `mnt=16`
G	`int4_sim`	2	0.88	n/a	long_ctx 1024 (Mistral), verifier exact
O	`kivi_offline`	0	0.000	n/a	LB NarrativeQA: `-!!!!...` vs prose
P	`int4_sim`	1	0.00	1	BFCL export-50: schema drift

Table 29: Historical release panel case studies. See §10 for full autopsy across 1,103 divergent cells.

17 Kernel Microbenchmark

Kernel microbenchmark only, *not* end-to-end inference speedups.
Source: `reports/phaseF_kernel_benchmark.json`.
Test configuration: shape $[1, 8, 512, 64]$, CUDA.

Table 30: Torch vs Triton kernel latency (ms). Ratio = torch / Triton. Verifier timing (evidence-plus): mean 3.85 s/cell, p90 5.18 s (diagnostic only).

Mode	torch (ms)	Triton (ms)	Ratio
`int8`	0.115	0.0706	1.63×
`int4`	0.3319	0.2162	1.54×
`block_sparse`	0.7622	0.7797	0.98×*

*`block_sparse` uses the torch execution backend only (not Triton-accelerated).

18 VeriCache: Closest Prior Art

VeriCache [8] is closer than a casual skim suggests. Both use compressed draft + full-KV verify/correct and target lossless greedy equivalence. **ExactKV must not claim** to invent that loop, VeriCache owns it for **lossless serving**.

Table 31: VeriCache vs ExactKV, complementary goals (ExactKV does *not* reproduce VeriCache).

	VeriCache	**ExactKV**
Goal	Serve faster, same outputs	Measure drift / first divergence
System	GPU compressed KV, offloaded full KV, staggering	Evaluation harness only
Metrics	Throughput, tiering	`first_divergence`, acceptance, failure
Reproduced?	n/a	**No**

19 Why ExactKV Still Matters After VeriCache

VeriCache uses verification to serve faster without changing outputs. ExactKV uses verification to measure exactly where and how compressors drift.

1. Compressor-agnostic crash-test on a fixed prompt panel.
2. Lossy-path first divergence (unverified `generate_lossy_greedy`).
3. Public diagnostic leaderboard with frozen cell definitions.
4. Explicit taxonomy: lossy drift vs acceptance vs `exactkv_failure`.

ExactKV does *not* replace VeriCache for deployment.

20 Related Work

Verification-based serving. VeriCache [8] is the closest prior art. Both systems use a compressed-KV draft path and a full-KV verify/correct loop. The distinction is purpose: VeriCache is a throughput-oriented *serving system* (scheduling, memory tiering, cross-resource staggering); ExactKV is a *diagnostic measurement framework* reporting where the unverified compressed path first diverges and how often. ExactKV does not reproduce VeriCache's system design and makes no claim to the draft+verify algorithm. Speculative decoding [3] and MagicDec [1] use draft/verify for throughput acceleration; ExactKV borrows the same semantics purely for measurement.

KV quantization methods. KVQuant [2] proposes non-uniform per-channel INT4 quantization targeting KV outlier heads. KIVI [6] proposes asymmetric 2-bit per-channel KV quantization with a CUDA kernel. Both directly motivate `int4_per_vec_sim`, which simulates per-vector granularity without a production kernel. The `kivi_offline` adapter (real KIVI quantizer math, no CUDA/Triton kernels) revealed 100% divergence in the current offline integration (Table 11) — a diagnostic, not a claim about KIVI's algorithmic accuracy. SnapKV [4] selects important KV entries by eviction; the `h2o_sim` family models the H2O Heavy Hitter Oracle policy [9] from the same compressor class and shows 100% LongBench divergence even at 75% retention (Section 8).

KV storage and streaming. CacheGen [5] compresses KV caches for network streaming and bandwidth reduction. LMCache [7] targets KV reuse and offloading across serving instances. Neither is designed to measure when or why the compressed path first diverges from the full-KV oracle.

Evaluation methodology. Standard benchmarks (LongBench, BFCL, HumanEval) measure task-level accuracy averaged over many tokens. ExactKV's first-divergence index (fdi) is a token-resolution diagnostic orthogonal to task accuracy: a cell can score perfectly on a task metric while accumulating 90% token-level drift (if the verifier corrects it). ExactKV uses official HF LongBench and BFCL datasets but reports *drift measurements*, not official benchmark scores.

21 Novelty and Claim Boundaries

Defensible (narrow): compressor-agnostic diagnostic/leaderboard for token-level drift and first-divergence under verifier semantics.

Not novel: compressed draft + full-KV verify (VeriCache), draft-verify for acceleration (speculative decoding), lossy-KV divergence observation (VeriCache problem statement).

ExactKV is *not* a production serving system and does *not* reproduce VeriCache. SpectralQuant is fallback/proxy, Shard is probe-first. Compression ratios are stored tensor byte ratios.

22 Future Work and Limitations

Completed: 144-cell evidence-plus (512/1024); 216-cell Llama-only external smoke (LongBench/RULER/BFCL/HumanEval bundled pilots); 144-cell MBPP smoke (both models); **640-cell `kivi_offline` panel** over real HF LongBench + MBPP subsets (see Table 11). `kivi_offline` uses real KIVI quantizer math (`models.utils_quant` simulate path; no CUDA/Triton kernels). The panel reveals catastrophic adapter-level drift (100% divergence, acceptance ≈ 0) and all cells corrected (`exactkv_failures=0`).

Future work — KIVI production path: validate `kivi_offline` against KIVI CUDA/Triton production serving path; validate byte-savings claim; integrate KVQuant and SnapKV. The offline simulate-path result is an adapter diagnostic, not a measurement of production KIVI quality.

Completed v2.9 work: BFCL validity panel v2.7 (1,200 cells, mnt=128/256, both models, `exactkv_failures=0`); H2O-style token-eviction panel v2.8 (800 cells, 5 eviction variants, both models); top-k logit autopsy across 1,103 divergent cells (`--store-top-k-logits`), revealing three mechanistically distinct failure modes. Total: **8,132 completed GPU cells**.

Remaining future work: expand BFCL validity to larger/more diverse prompts; expand MBPP with safe pass@1/syntax validation; broader downstream metrics beyond BFCL (including LongBench answer-overlap); RULER 12K+; HELMET; InfiniteBench 100K+; broader scaling curves (12K+, 16K+ context; more compressors); faithful production KIVI/KVQuant/SnapKV integrations.

23 Conclusion

VeriCache shows how to **serve** with compressed KV without changing outputs. ExactKV shows **where and why compressors drift** on the lossy path.

ExactKV's strongest supported claim is that **KV-cache drift is governed jointly by task type, generation length, compressor class, and quantization granularity** — and that verifier-mediated decoding preserves full-KV greedy equivalence (`exactkv_failures=0` throughout).

Across **8,132 completed GPU cells** with `exactkv_failures=0` throughout, ExactKV establishes four empirical findings:

1. **Task type dominates drift.** `int4_sim` diverges 6% on Python code (MBPP), 11% on BFCL short tool-calling, 50% on BFCL long tool-calling, and 90% on HF LongBench reading/summarization.

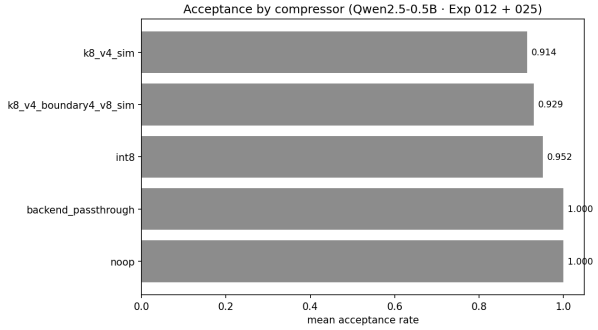
2. **Generation length is the dominant within-task driver.** On BFCL, `int4_sim` divergence scales from 9% (mnt=16) to 62% (mnt=256) — a 7× increase. `int8` stays near-zero throughout.

3. **Compressor class determines failure mode.** Logit autopsy over 1,103 divergent cells reveals: (a) near-tie noise (int8, mean rank 2.4, fdi=22); (b) distribution shift (int4_sim, rank 3.5, fdi=8); (c) attention destruction (H2O-style, rank 6.7, fdi=1).

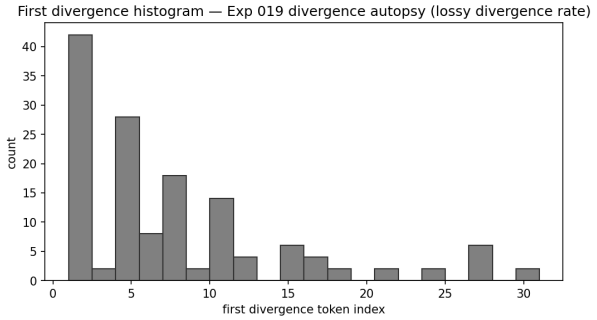
4. **Downstream task validity is fully preserved by ExactKV.** Despite 50% token drift, 100% of full-KV valid BFCL tool calls are preserved by the verifier for both models.

Two new compressors extend the compression curve: `int6_sim` (6-bit per-tensor) and `int4_per_vec_sim` (4-bit per-vector, KIVI/KVQuant-style). The v3.0 GPU panel (both models, 1,568 cells, `exactkv_failures=0`) validates both as non-catastrophic: 0% on BFCL/MBPP; 37–47% (int6) and 56–57% (int4 per-vector) on LongBench. Per-vector granularity eliminates drift on structured tasks but 4-bit resolution still accumulates at 8K context.

ExactKV's contribution is not a new compressor. It is a measurement infrastructure that makes KV-cache drift **observable, attributable, and reproducible** — and a verifier mechanism that corrects it with zero exactness failures across all 8,132 tested cells.



(a) Acceptance by compressor.



(b) First-divergence distribution.

Figure 1: Release benchmark visuals (`reports/scale_7b/`).

A All Completed Panels — Benchmark Card

Panel	**Models**	**Ctx (K)**	**Cells**	**EKV fail**
Headline release	Llama+Mistral	0.5–2	1,500	0
Evidence-plus	Llama+Mistral	0.5,1	144	0
Ext. smoke: LongBench	Llama only	2,4	72	0
Ext. smoke: RULER 2K–4K	Llama only	2,4	48	0
Ext. smoke: RULER 8K	Llama only	8	24	0
Ext. smoke: BFCL pilot	Llama only	1,2	48	0
Ext. smoke: HumanEval	Llama only	1,2	24	0
MBPP code-drift smoke	Llama+Mistral	0.5,1	144	0
BFCL export-50 tool-call	Llama+Mistral	1,2	1,200	0
KIVI offline adapter	Llama only	2,4	640	0
HF LongBench v2.6	Llama+Mistral	2,4,8	720	0
BFCL validity v2.7	Llama+Mistral	1,2	1,200	0
H2O token-eviction v2.8	Llama+Mistral	2,4	800	0
v3.0 int6/int4pv (Mistral)	Mistral	2,4,8+	784	0
v3.0 int6/int4pv (Llama)	Llama	2,4,8+	784	0
Subtotal (pre-v3.0)			6,564	
Subtotal (v3.0)			1,568	
Grand total			**8,132**	**0**

Table 32: All completed ExactKV GPU panels as of v3.0. `exactkv_failures=0` throughout. Reconciliation: $6{,}564 + 784 + 784 = 8{,}132$. Each v3.0 row = one model × four compressors × three task families (784 cells). Source: `reports/external_panels/v30/`. Not official benchmark scores.

B Divergence Autopsy: Extended Details

See Section 10 for the full mechanistic analysis and logit autopsy table. This appendix section provides the methodology reference: at each divergence position, ExactKV records the top-5 full-KV token probabilities and the actual lossy-chosen token. Metrics: top-1 flip (different argmax), near-tie (full-KV margin < 0.05), mean lossy rank (where the lossy-chosen token ranks under full-KV distribution), and median first-divergence index (fdi). Script: `scripts/analyze_logit_margins.py`.

Panels with stored top-k logits: v2.6 LongBench, v2.7 BFCL validity, v2.8 H2O-style.

C Reproducibility

Artifact verification (no GPU):

```
python3 scripts/exactkv_repro.py --reports-only
python3 scripts/exactkv_repro.py --release-check
bash scripts/build_paper_pdf.sh   # requires: tectonic
```

Headline panel (GPU):

```
python3 scripts/run_phase_a_scale_benchmark.py --device cuda
```

Evidence-plus panel (GPU):

```
python3 scripts/run_evidence_plus_panel.py --device cuda --dtype float16
```

External smoke panels — Llama-only (GPU, 216 cells):

```
bash scripts/run_external_gpu_workflow.sh
python3 scripts/build_external_analysis_pack.py
python3 scripts/build_external_panel_summary.py --write-readme
python3 scripts/validate_external_panel_artifacts.py \
    --input reports/external_panels
```

MBPP code-drift smoke panel (GPU, 144 cells, both models):

```
python3 scripts/run_external_panel.py \
  --family mbpp --device cuda --dtype float16 \
  --max-prompts 6 --context-buckets 512,1024 \
  --max-new-tokens 16,32 \
  --output-json reports/external_panels/mbpp_gpu_raw.json
python3 scripts/validate_external_panel_artifacts.py \
    --input reports/external_panels
```

BFCL export-50 tool-call drift panel (GPU, 1200 cells, both models):

```
python3 scripts/run_external_panel.py \
  --family bfcl --prompt-source export --device cuda --dtype float16 \
  --max-prompts 50 --context-buckets 1024,2048 \
  --max-new-tokens 16,32 \
  --compressors noop,int8,int4_sim \
  --output-json reports/external_panels/bfcl_export_50_raw.json
# Artifact: reports/external_panels/bfcl_export_50_raw.json (13.5 MB)
# Note: max_new_tokens={16,32} measures drift, not JSON completeness.
#       For tool-call validity, use max_new_tokens={128,256}.
```

KIVI offline compressor panel (GPU, 640 cells — requires KIVI `PYTHONPATH`):

```
# Requires: git clone https://github.com/jy-yuan/KIVI /tmp/kivi_research
#           pip install datasets>=2.0
export PYTHONPATH=/tmp/kivi_research
bash scripts/run_kivi_external_panel.sh
# Artifacts: reports/external_panels/kivi_longbench_hf_raw.json
#            reports/external_panels/kivi_mbpp_hf_raw.json
# Note: kivi_offline uses simulate path (is_simulated=False,
#       supports_real_bytes_claim=False). Not production KIVI serving.
```

H2O-style token-eviction panel (GPU, 800 cells, v2.8):

```
python3 scripts/run_external_panel.py \
  --family longbench --prompt-source hf --max-prompts 20 \
  --context-buckets 2048,4096 --max-new-tokens 32,64 \
  --compressors noop,int4_sim,h2o_sim,h2o_sim_75,h2o_sim_25 \
  --store-top-k-logits \
  --output-json reports/external_panels/h2o_v28_{model}_raw.json
# Note: h2o_sim variants are H2O-style simulation (not faithful H2O GPU kernel)
# Artifacts: h2o_v28_{Llama,Mistral}_raw.json, h2o_v28_merged_raw.json
```

Top-k logit autopsy (post-hoc, requires –store-top-k-logits panels):

```
python3 scripts/analyze_logit_margins.py \
  --inputs reports/external_panels/hf_longbench_v26_merged_raw.json \
           reports/external_panels/bfcl_validity_v27_merged_raw.json \
           reports/external_panels/h2o_v28_merged_raw.json \
  --output reports/external_panels/logit_autopsy_summary.json
# Produces Table 19 metrics across 1,103 divergent cells
```

Sources of truth (all under `reports/`): `scale_7b/raw.json` (1,500, headline), `evidence_plus/raw.json` (144, evidence-plus), `external_panels/summary_all.json` (216, smoke), `external_panels/mbpp_gpu_raw.json` (144, MBPP), `external_panels/bfcl_export_50_raw.json` (1,200, BFCL export-50), `external_panels/kivi_longbench_hf_raw.json` (640, KIVI offline), `external_panels/hf_longbench_v26_merged_raw.json` (720, v2.6), `external_panels/bfcl_validity_v27_merged_raw.json` (1,200, v2.7), `external_panels/h2o_v28_merged_raw.json` (800, v2.8). Total: **8,132 completed GPU cells**, `exactkv_failures=0`.

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